

Storage Connection Fault Simulation Report

Test Procedure and Observations

1. Initial System Check (Drive Connected)

Action:

- Powered on the test system with the storage drive properly connected.

Observed Results:

- System completed POST (Power-On Self-Test).
- BIOS/UEFI detected the storage device.
- Boot drive appeared in the boot priority list.
- Operating system loaded normally.

Example BIOS Indicator:

- Storage device listed under SATA/NVMe information:
 - Example: "SATA SSD: Detected"
 - Example: "NVMe Drive: Present"
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2. Simulated Drive Disconnection

Action:

- Powered off the system.
- Disconnected the SATA data cable/power cable or removed the NVMe drive from its slot (simulation).

Observed Results:

- System could no longer locate the operating system drive.

Possible Error Messages:

- "No boot device found"
- "Boot device not detected"
- "Insert proper boot device and press any key"
- "Operating system not found"

BIOS Indicators:

- Storage drive missing from the SATA/NVMe device list.
 - Boot option for the drive no longer available.
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3. Drive Reconnection

Action:

- Powered off the system again.
- Reconnected the SATA cable/power connector or reseated the NVMe drive.
- Ensured connections were secure.

Observed Results:

- BIOS/UEFI detected the storage device again.
- Drive returned to the boot menu.
- System was able to locate the operating system.

Example BIOS Indicator:

- Drive listed again under storage information:
 - "SATA SSD: Detected"
 - "NVMe SSD: Present"
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4. Final System Verification

For a **PC that randomly restarts**, I would check **RAM first**, followed by **GPU**, then **storage**. The reason is that random restarts are often caused by unstable memory, overheating, power-related faults, or hardware errors that trigger system crashes. Among the listed components, **RAM is a strong first suspect because faulty or unstable memory can cause unpredictable system behavior, including sudden reboots, freezes, and crashes without obvious warning.**

1. First Component to Check: RAM (Memory)

Reasoning

- Faulty RAM can corrupt data being used by the operating system, causing:
 - Random restarts
 - Blue Screen of Death (BSOD) errors
 - Application crashes
 - Failure to boot
- RAM problems are relatively easy to isolate by testing modules individually.
- Storage failures usually cause slow performance, file errors, or boot problems rather than immediate random restarts.
- GPU problems often appear during gaming or graphics-heavy workloads, while RAM faults can happen at any time.

RAM Diagnostic Steps

☐ 1. Check physical installation

- Shut down the PC and disconnect power.
- Open the case and verify that:
 - RAM sticks are fully seated.
 - Locking clips are properly engaged.
 - There is no visible dust or damage around the RAM slots.

☐ 2. Reseat the RAM

- Remove each RAM module.
- Reinstall it firmly into the correct slot.
- Test whether the random restarts continue.

☐ 3. Test one RAM stick at a time

- Remove all but one RAM module.
- Boot the system and test stability.
- Repeat with each module to identify a faulty stick.

☐ 4. Run a memory diagnostic

- Use a tool such as MemTest86 or the built-in Windows Memory Diagnostic.
- Look for:
 - Memory errors
 - Failed test patterns
 - Address errors

☐ 5. Check BIOS memory settings

- Enter BIOS/UEFI.
 - Verify:
 - RAM is detected correctly.
 - Memory speed matches specifications.
 - Disable overclocking profiles such as XMP/EXPO temporarily to test stability.
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2. Second Component to Check: GPU

Reasoning

A faulty GPU can cause restarts, especially when the system is under load:

- Gaming
- 3D rendering
- Video processing

GPU Diagnostic Steps

☐ Check that the GPU is:

- Properly seated in the PCIe slot.

- Connected to the correct PCIe power cables.

☐ Monitor GPU temperatures:

- Use a monitoring tool.
- Check whether overheating occurs before restart.

☐ Update or reinstall GPU drivers:

- Remove corrupted drivers if necessary.
- Install the latest stable driver from the GPU manufacturer.

☐ Run a GPU stress test:

- Check whether the restart occurs only during heavy graphics workloads.

☐ Test with another GPU (if available):

- A stable replacement GPU can confirm whether the original card is faulty.
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3. Third Component to Check: Storage (SSD/HDD)

Reasoning

Storage problems usually cause:

- Boot failures
- Corrupted files
- Slow loading
- Application errors

They are less commonly the direct cause of sudden restarts, but a failing drive or corrupted system files can contribute.

Storage Diagnostic Steps

☐ Check drive connections:

- Ensure SATA data and power cables are secure.
- Reseat an NVMe drive if applicable.

☐ Check BIOS detection:

- Confirm the drive appears correctly.

☐ Check drive health:

- Use SMART monitoring tools.
- Look for:
 - Bad sectors
 - High error counts
 - Drive failure warnings

☐ Run file-system checks:

- Scan for corrupted operating system files.
- Repair disk errors if detected.

☐ Test with another drive:

- Install a known-good drive and check whether restarts continue.